Last update: September 2025

Spontaneous abortion (miscarriage)

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Introduction

There is a conceptual difference regarding the word “abortion”. A distinction should be made between a spontaneous abortion due to natural causes (“miscarriage”) and an abortion brought about by medical intervention (“intentional termination of pregnancy” or “induced abortion”; see below).

Cave

In English, the term “abortion” can refer to either a miscarriage or an intentional termination of pregnancy. In fact, the term is not clearly defined. Ideally, more specific terms should be used: “spontaneous abortion” for miscarriage and “induced abortion” or “therapeutic abortion” for intentional termination of pregnancy.

> To avoid confusion, it is advisable to consistently refer to intentional termination of pregnancy as “induced abortion”.

From a practical point of view, spontaneous abortions are classified as either early or late. Early miscarriages are usually caused by defective embryonic development. Late miscarriages, on the other hand, often require further investigation because they are frequently caused by maternal factors that may pose a risk of recurrence.

Due to recent advances in neonatology, the lower limit of fetal viability has shifted. Consequently, events occurring after the 22nd week of gestation (WG) are no longer referred to as spontaneous abortions but as intrauterine fetal deaths (IUFDs). Partus immaturus refers to the birth of an immature, nonviable child. Previously, it was defined as occurring between 22 1/7 and 23 0/7 WG, but as mentioned, the upper limit is now lower.

Definition

Spontaneous abortion = miscarriage

- Premature spontaneous termination of pregnancy with or without expulsion of the dead embryo or fetus
- with a birth weight of < 500 g or
- if the weight is unknown, at a gestational age of < 22 0/7 WG

Classification of spontaneous abortions according to the time of occurrence

- Early miscarriage: up to 12 0/7 WG
- Late miscarriage: 12 1/7 to 21 6/7 WG

Nomenclature of spontaneous abortions according to clinical situation

- Abortus imminens (threatened abortion)
- Abortus incipiens (incipient, inevitable abortion)
- Abortus incompletus (incomplete abortion)
- Abortus completus (complete abortion)
- Missed abortion: A pregnancy that is no longer viable and is in utero without symptoms (such as bleeding, etc.).
- Abortus febrilis (febrile to septic abortion)
- Habitual abortion (at least 3 consecutive spontaneous abortions with the same partner in a row up to 20 WG; frequency 1 to 3% of all couples of reproductive age): The causes here are usually multifactorial and often cannot be determined individually

Also to be mentioned is the wind egg = blighted ovum. This is a molar trophoblast without embryonic development, i.e., a germ malformation in which the embryoblast does not form, resulting in only a chorionic cavity without an embryo or yolk sac. Caveat: Not to be confused with a hydatidiform mole (cf. also EGONE script G12B “Gestational trophoblastic diseases”), which is classified as a villous trophoblastic tumor!

Epidemiology

Spontaneous abortion is the most common complication in early pregnancy. Up to 30 % of clinically confirmed pregnancies before the 20th WG end in spontaneous abortion. In 80 % of cases, this occurs during the first 12 WG. The overall risk of spontaneous abortion after the 15th WG is low, around 0,6 % for chromosomally and structurally normal fetuses, but it varies with maternal age and ethnicity.

The loss of subclinical or undetected pregnancies is significantly higher. Nowadays, it is assumed that 30 to 60 % of all pregnancies end in spontaneous abortion.

Under hCG monitoring, it has been shown that the overall rate of pregnancy loss is 31 %, of which 70 % occurred before the pregnancy was clinically detectable.

Regarding preimplantation losses, it is currently assumed that 50% of fertilized oocytes do not result in a live birth.

Early miscarriages are more common (80 %) than late miscarriages. Spontaneous abortions show a clear predominance for male fetuses, with a ratio of 3:2.

Risk factors

Age

- 20 to 29 years old: 12 %
- > 45 years old: 65 %
- Gradual increase in between

Previous spontaneous abortion

The risk of a subsequent spontaneous abortion is 20 % after one previous spontaneous abortion, 30 % after two, and approximately 40 % after three or more (consecutively). For women who had a successful first pregnancy, the risk of a spontaneous abortion is only 5 %.

Smoking

Heavy smokers (more than 10 cigarettes per day) have a higher risk of experiencing a spontaneous abortion.

Alcohol

Studies are inconsistent in this regard. As a general rule, alcohol should be avoided during pregnancy due to its teratogenicity.

Poorly controlled maternal disease

This specifically applies to diabetes and thyroid disorders (both hyperthyroidism and hypothyroidism). There is no known increased risk of spontaneous abortion in individuals with euglycemic and euthyroid metabolic status. In addition, untreated celiac disease also appears to increase the risk of spontaneous abortion.

Cocaine

Cocaine is associated with an increased risk of spontaneous abortion, but is primarily known for causing premature birth.

Caffeine

The impact of caffeine is unclear. Up to 3 cups per day are considered safe.

Maternal weight

Overweight (BMI >25 kg/m²) is associated with a markedly increased risk of spontaneous abortion. Both underweight and overweight are also associated with increased infertility.

Subchorionic hematoma

According to studies, a subchorionic hematoma leads to a higher spontaneous abortion rate if it occupies more than 25 % of the gestational sac volume.

Etiology

Etiology of early miscarriages

- In over 50 % of cases, the cause is a chromosomal aberration.
- The rate of early miscarriages increases with maternal and paternal age.
- The harmful influence of noxae, especially nicotine, has been proven to increase the spontaneous abortion rate.
- In cases of early miscarriage (< 6 0/7 WG), corpus luteum insufficiency may be a contributing factor.

Etiology of late miscarriages

- Cervical insufficiency (risk factors: history of cervical trauma, infection, history of conization, history of cervical insufficiency) is often involved. The cervix can no longer adequately fulfill its supporting and closing functions, leading to an (often asymptomatic) opening of the cervix that initially goes unnoticed.
- Mechanical insufficiency is often accompanied by an ascending infection in the form of chorioamnionitis.
- Anatomical cause (limited increase in uterine volume): uterine malformations, myoma.

Etiology of habitual abortion

Only in cases of habitual abortion, defined as ≥ 3 consecutive spontaneous abortions before the 20th WG with the same partner, are further etiological investigations indicated.

The following diagnostic procedures may be used to determine the causes of habitual abortion:

- BMI, medical history of noxious exposure
- Hormone analyses (thyroid, PCOS, prolactin, sugar)
- Exclusion of infections with blood samples and cervical smears, endometrial biopsy if necessary to rule out chronic endometritis
- (Auto)immunoserology (antiphospholipid syndrome)
- Exclusion of uterine malformations by sonography, hysteroscopy, hysterosalpingography (Ashermann's syndrome, polyps, myoma)
- Chromosome analysis of the couple (genetic counseling necessary)
- Karyotyping of the abortion material

- Coagulation testing (general thrombophilia screening is not recommended)

List of maternal causes of habitual abortion

Genital factors/abnormalities:

- Malformations: uterine duplication, intrauterine synechiae
- Submucosal myomas
- Cervical insufficiency: possibly previous operations (conization), connective tissue weakness
- Endometrial insufficiency: endocrine disorder, atrophy after curettage
- Infection: cervicitis, endometritis
- Hypermotility (psychovegetative disorder, fever)
- Adenomyosis/endometriosis

Extragenital factors/abnormalities:

- Noxae: alcohol, nicotine
- Immunological (blood group antibodies, antiphospholipid syndrome)
- Endocrine disorders (diabetes mellitus, thyroid disorders)
- Viral and bacterial infections (fever-induced hypermobility of the uterus, infectious or toxic damage to the fetus)
- Consumptive diseases

List of fetoplacental causes of habitual abortion

- Chromosomal aberrations
- Trophoblastic dysfunction (endocrine insufficiency)

Clinical presentation and diagnosis

The symptoms of spontaneous abortion vary greatly. A spontaneous abortion may go completely unnoticed or may be accompanied by bleeding.

- In rare cases, bleeding can be extensive and even life-threatening. In other cases, only minor spotting occurs.
- Occasionally, abdominal discomfort (cramping or pain) may also be present, often described as “heavier menstrual cramps”.
- Particular caution is advised in cases of fever and purulent discharge (--> an indication of an infectious complication of the abortion and thus a medical emergency)!

Early miscarriages (up to approximately 6 WG) are often subclinical and are mistaken for a delayed menstrual period. A positive pregnancy test provides an indication! These very early miscarriages do not require further treatment. Only 20 % of these are perceived by the pregnant woman as a miscarriage.

The diagnosis of spontaneous abortion is made or verified using ultrasound.

Confirmation of the miscarriage diagnosis

In early pregnancy, especially without a prior ultrasound to determine the gestational age, diagnosing a spontaneous abortion can sometimes be difficult. Reliable signs of miscarriage are:

- Crown-rump length ≥7 mm without positive cardiac activity
- Gestational sac with a diameter ≥25 mm without embryonic development
- Absent embryo with positive cardiac activity 2 weeks after an ultrasound in which a gestational sac but no yolk sac was visible
- Absent embryo with positive cardiac activity ≥11 days after a gestational sac with yolk sac was detected

Clinical engrams for the individual forms of spontaneous abortion

Abortus imminens

Threatened abortion manifests as (light) vaginal bleeding with an otherwise intact pregnancy.

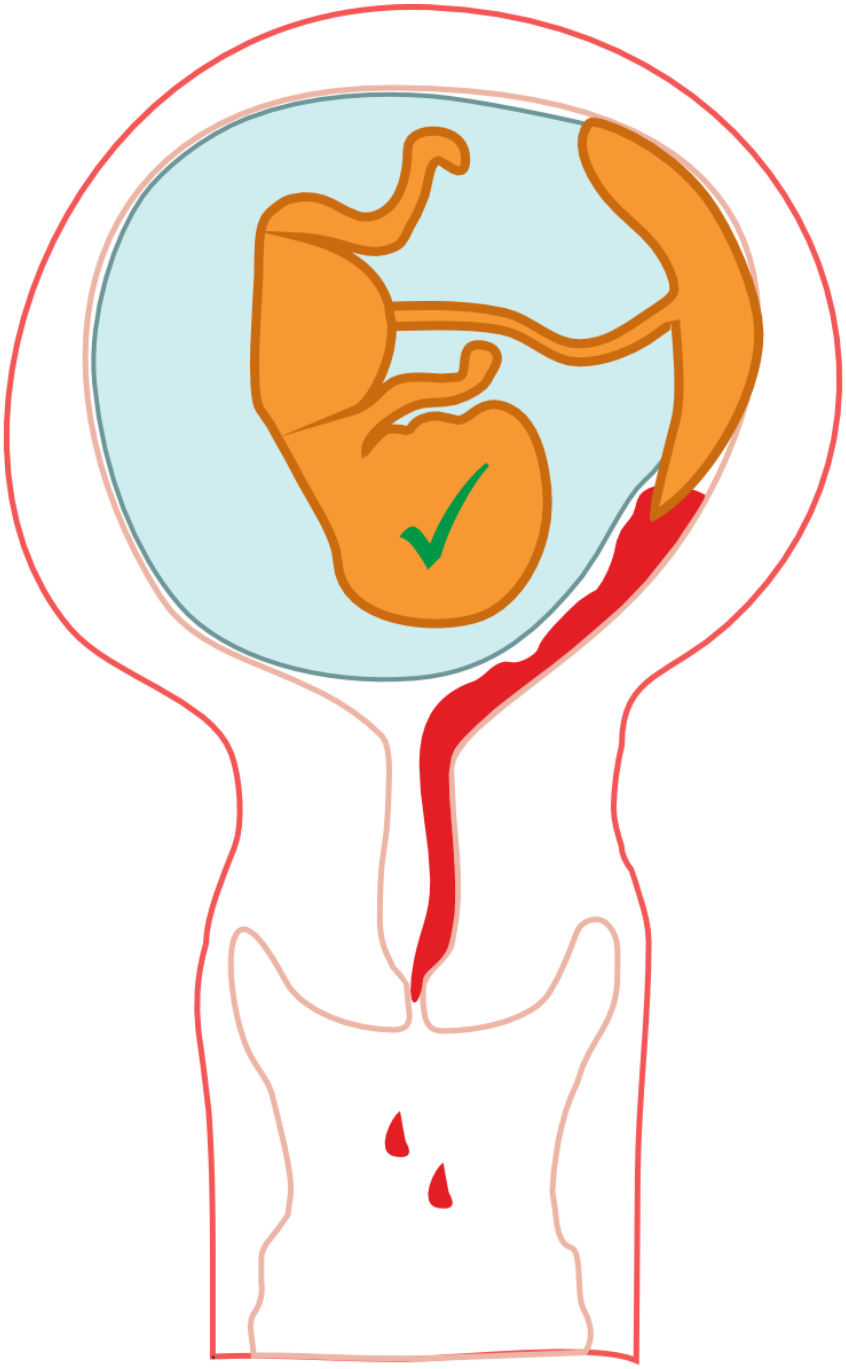

Figure 1: Abortus imminens.

The cervix is preserved. The clinically decisive factor is:

- The cervix is still closed.

Vaginal examinations should be kept to a minimum. Ultrasound may reveal a retroplacental hematoma, the extent of which is a major prognostic factor.

Abortus incipiens

The abortion process has begun. In most cases, the patient experiences a menstrual-like bleeding. The pregnancy has been irreversibly disrupted and the abortion can no longer be stopped.

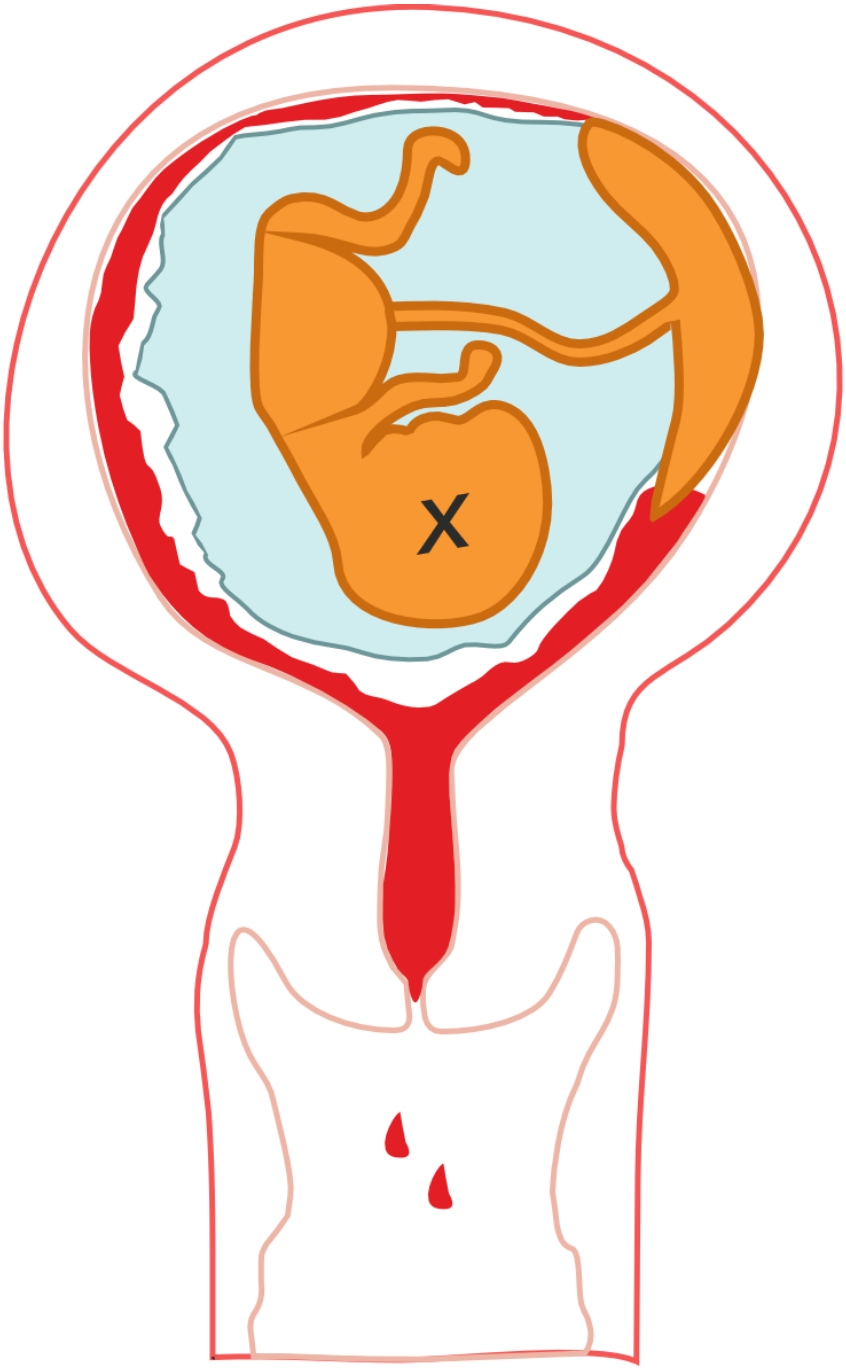

Figure 2: Abortus incipiens.

The cervix is opened. The bleeding is usually accompanied by lower abdominal pain and leads to an incomplete or complete abortion.

Incomplete and complete abortion

In incomplete abortion, part of the pregnancy material has been expelled. However, the uterine cavity is not empty. In complete abortion, all of the pregnancy material has been expelled.

Missed abortion

In the case of a missed abortion, there are generally no symptoms. It is mainly characterized by the “silent” termination of the pregnancy. It is often diagnosed incidentally during a routine pregnancy check-up. Ultrasound shows no cardiac activity and the crown-rump length does not correspond to the gestational age.

Blighted ovum

The clinical presentation corresponds to that of a missed abortion. Ultrasound reveals a gestational sac without a yolk sac or embryo.

Abortus febrilis

After an unprofessional attempt to induce an abortion, bacterial ascension can lead to an infection of the decidua and myometrium (endomyometritis), which can spread throughout the body. Possible consequences include parametritis, peritonitis, and septicemia with endocarditis. A septic abortion can progress to life-threatening septic shock with renal failure, respiratory failure (ARDS), and disseminated intravascular coagulation (DIC).

Differential diagnosis

Differential diagnosis of vaginal bleeding in early pregnancy

- in causal connection with pregnancy: the various forms of spontaneous abortion, ectopic pregnancy
- not in causal connection with pregnancy: polyp in the cervical canal (CC), infections, ectopy (“ectopic bleeding”)

Differential diagnosis of pain in early pregnancy

- in causal connection with pregnancy: the various forms of spontaneous abortion, ectopic pregnancy, stretching pain (uterus)
- not in causal connection with pregnancy: ovarian cysts (with/without bleeding, rupture, torsion), entire range of internal medicine and surgery conditions (appendicitis*, renal colic, constipation, etc.)

*Note: Appendicitis can manifest atypically throughout pregnancy and should always be ruled out in cases of pain and positive inflammation parameters.

Treatment of threatened abortion (abortus imminens)

Important: If the woman is rhesus negative, anti-D prophylaxis must be administered for all forms of spontaneous abortion beyond 7 4/7 WG!

Abortus imminens

The “treatment” for threatened abortion consists of psychophysical relief. Generous sick leave is recommended. Sexual intercourse should be avoided. A meta-analysis from 2021 involving 5,682 patients concluded that the administration of vaginal progesterone in cases of bleeding in early pregnancy and recurrent abortions can have a positive effect on the survival of the child. Vaginal progesterone administration is highly unlikely to cause adverse events.

Management of early miscarriages

For all forms of early miscarriage, there are three possible options for hemodynamically stable women without heavy bleeding or fever, which should be explained to the patient. In the case of a febrile abortion, immediate antibiotic treatment and termination of pregnancy by curettage are indicated. From 7+4 WG onwards, 300 mcg of anti-D immunoglobulin should be administered in the case of a negative blood group.

1. Wait-and-see management
2. Drug-based management
3. Surgical management

Wait-and-see management

A wait-and-see approach is an alternative for women before the 13th WG who are in stable clinical condition with no signs of infection. According to the British NICE guideline, this is even recommended as the first-line treatment for the first two weeks after diagnosis. Compared to the surgical option, there is an increased risk of incomplete expulsion. There may be more frequent bleeding and a consequent need for curettage. The patient should be well informed about this. Most expulsions occur within the first 2 weeks. After 6 weeks, 81 % of patients have had a complete abortion, which, according to randomized studies, corresponds to the success rate of the drug-based option. There are no significant differences between the three options in terms of infection risk. Follow-up checks every 2 weeks are generally recommended. If bleeding does not occur within 2 weeks, switching to the drug-based or surgical option can be discussed. There is no consensus in the guidelines regarding the definition of an incomplete abortion. In general, clinical findings (e.g., persistent bleeding) should be taken into account. According to an ACOG Practice Bulletin, an endometrial thickness of > 30 mm is considered an incomplete abortion. However, according to a Dutch review, most studies used a cut-off of > 15 mm. In some practices/countries, clinical monitoring is not performed if there has been a history of bleeding and a β-hCG test was negative 4 weeks after the initial presentation.

Drug therapy with misoprostol

In drug-induced abortions, administering mifepristone (Mifegyne, a progesterone receptor antagonist used off-label) before misoprostol (a prostaglandin E1 analog) has proven effective. A Dutch double-blind, placebo-controlled study showed that 79,1 % of participants who received Mifegyne had a complete abortion, compared to 58,7 % in the placebo group.

Procedure

Administration of 200 mg Mifegyne p.o., followed by 600 to 800 mcg misoprostol vaginally or p.o. 36 to 48 hours after taking Mifegyne. An ultrasound control should be performed 7 to 14 days later. If residual material is present, a repeat dose of misoprostol may be considered before opting for surgery. Typical side effects of the drugs include lower abdominal pain, nausea, vomiting, headache, dizziness, and diarrhea

Contraindications for the wait-and-see and drug-based approaches

- Low hemoglobin/circulatory instability
- Coagulation disorders
- Lack of compliance
- Rejection by the patient
- Suspected trophoblastic disease
- Suspected ectopic pregnancy
- Contraindications for misoprostol: glaucoma, sickle cell anemia, epilepsy, mitral stenosis
- Contraindications for mifepristone (Mifegyne): chronic corticosteroid use, adrenal disorders

Abortion curettage

Whenever possible, curettage should be performed using suction rather than spoon. The procedure is highly successful, with only 4 % of patients requiring further surgery. However, the incidence of recurrent spontaneous abortions after curettage is significantly higher than after drug-based treatment. Furthermore, there is an increased risk of Asherman's syndrome (especially with spoon curettage). In addition, the general risks of surgery, such as perforation with injury to neighboring organs, bleeding, and thrombosis, should not be neglected.

Treatment of septic abortion

If septic abortion is suspected (fever, elevated white blood cell count, residual material visible on ultrasound), the following procedure is recommended:

- Stabilization of the patient
- Collection of swabs
- Empirical broad-spectrum antibiotic therapy (later, pathogen-specific)
- Provision of erythrocyte concentrates
- Termination of pregnancy by curettage as quickly as possible

The most common pathogens are anaerobic bacteria, but also: coliform bacteria, group A streptococci, Haemophilus influenzae, Campylobacter jejuni. Treatment should be initiated with broad-spectrum antibiotics (e.g., clindamycin + gentamicin +/− ampicillin) appropriate for the resistance profile.
Caution: If Clostridium sepsis is suspected (crackling of tissues during gynecological examination, or radiographic suspicion of air in uterine tissue), an emergency hysterectomy may be indicated.

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